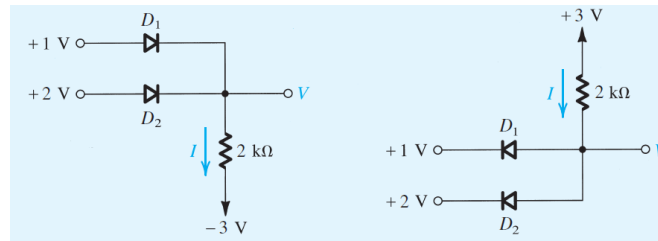


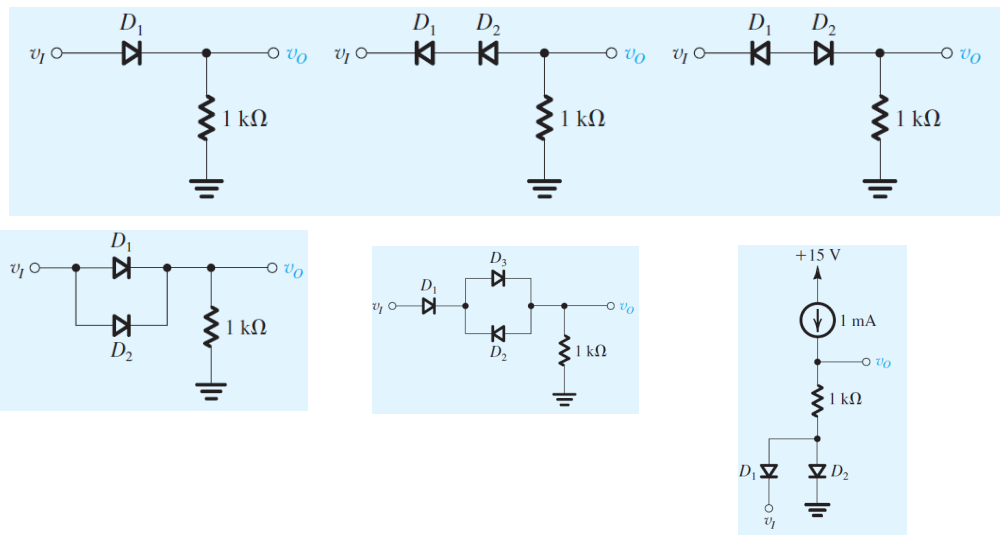
Homework 7

Topic: Diodes I

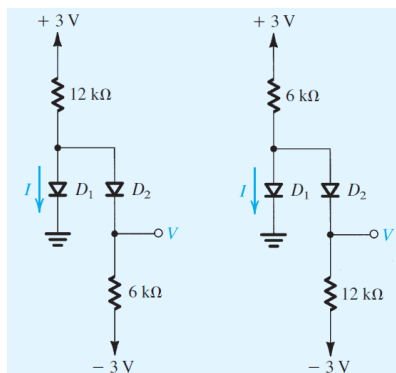
1. For the circuits shown below using ideal diodes, find the values of the labeled voltages and currents.



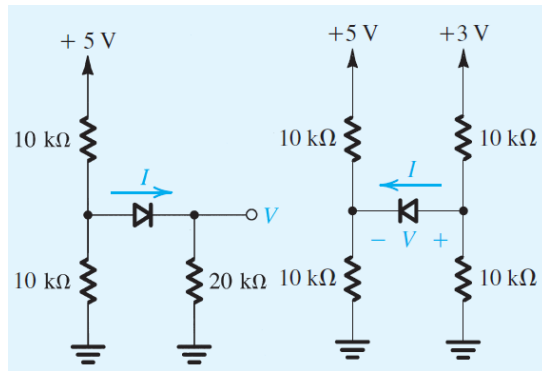
2. In each of the ideal-diode circuits shown below, v_i is a 1-kHz, 5-V peak sine wave. Sketch the waveform resulting at v_o . What are its positive and negative peak values?



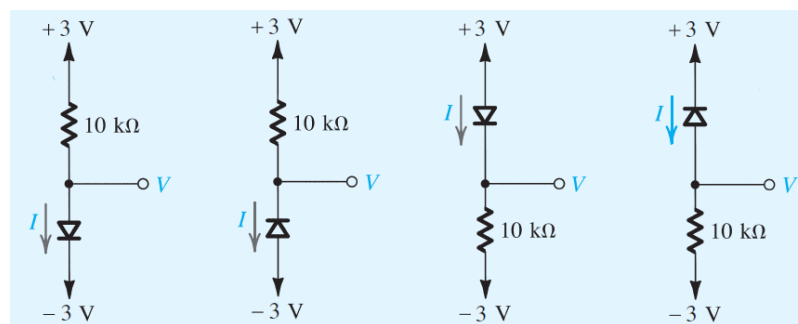
3. Assuming that the diodes in the circuits below are ideal, find the values of the labeled voltages and currents.



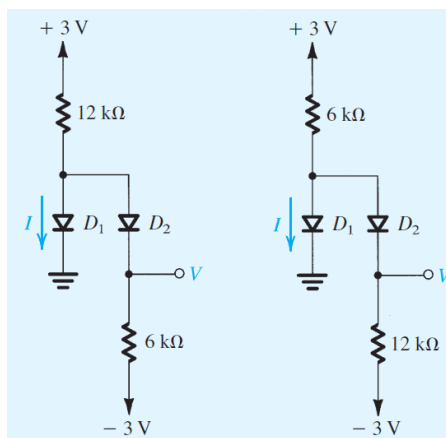
4. Assuming that the diodes in the circuits below are ideal, utilize Thevenin's theorem to simplify the circuits and thus find the values of the labeled currents and voltages.



5. For the circuits shown below, using the constant-voltage-drop ($V_D = 0.7$ V) diode model, find the voltages and currents indicated.



6. For the circuits below, using the constant-voltage-drop ($V_D = 0.7$ V) diode model, find the values of the labeled currents and voltages.



7. Repeat problems 3 and 4 using a piecewise linear diode model with $V_f = 0.6$ V, $r_d = 1$ k Ω , and no reverse bias current.